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Corrigendum: Observation-based solar and wind power capacity factors and power densities (2018 *Environ. Res. Lett.* [13 104008](#))

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Corrigendum abstract

An error in the estimate of wind plant area led us to underestimate wind power densities by about 40%. The error was our incorrect specification of the geometric projection in the calculation of the area of Voroni polygons in our GIS software. The severity of this error increased with latitude so errors were smaller in Texas than Montana. Our method used area to filter out plants with installed capacity densities $<0.1 \text{ MW}_i \text{ km}^{-2}$, a step that generally removes plants with very small numbers of turbines, for which our Voroni method produces overly-large areas. Because the areas changed, the sample set also changed when the error was fixed. Finding the error motivated us to both provide a more detailed description of the method in the supplemental information is available online at stacks.iop.org/ERL/14/079501/mmedia and make data publicly available in an Addendum.

The average wind power density changed to $0.90 \text{ W}_e \text{ m}^{-2}$ (from $0.50 \text{ W}_e \text{ m}^{-2}$) and the average installed capacity density changed to $2.8 \text{ MW}_i \text{ km}^{-2}$ (from $1.5 \text{ MW}_i \text{ km}^{-2}$). Yet while we are embarrassed to have made an error, these corrections do not affect the overall conclusions of the paper. Specifically: (a) wind plants with the largest areas have the lowest power densities; (b) wind capacity factors are increasing, and that increase is associated with a decrease in installed capacity densities, so power densities are stable or declining; and, (c) the observed average power densities are consistent with prior estimates that use physically-based models of turbine-atmosphere interaction and are inconsistent with many wind resource estimates that implicitly ignore these interactions. Corrections do change figures 3–7 and table 1, as well as text citing or comparing previously incorrect numbers. Paragraphs which required amendments are included below, with corrected numbers and text identifiable as bold-under-scored text. We apologize for the inconvenience.

Power density is the rate of energy generation per unit of land surface area occupied by an energy system. The power density of low-carbon energy sources will play an important role in mediating the environmental consequences of energy system decarbonisation as the world transitions away from high power-density fossil fuels. All else equal, lower power densities mean larger land and environmental footprints. The power density of solar and wind power remain surprisingly uncertain: estimates of *realizable* generation rates per unit area for wind and solar power span $0.3\text{--}47 \text{ W}_e \text{ m}^{-2}$ and $10\text{--}120 \text{ W}_e \text{ m}^{-2}$ respectively. We refine this range using US data from 1984 to 2016. We estimate wind power density from primary data, and solar power density from primary plant-level data and prior datasets on capacity density. The mean power density of **430** onshore wind power plants in 2016 was **0.90** $\text{W}_e \text{ m}^{-2}$. Wind plants with the largest areas have the lowest power densities. Wind power capacity factors are increasing, but that increase is associated with a decrease in capacity densities, so power densities are stable or declining. If wind power expands away from the best locations and the areas of wind power plants keep increasing, it seems likely that wind's power density will decrease as total wind generation increases. The mean 2016 power density of **1047** solar power plants was **5.7** $\text{W}_e \text{ m}^{-2}$. Solar capacity factors and (likely) power densities are increasing with time driven, in part, by improved panel efficiencies. Wind power has a **6**-fold lower power density than solar, but wind power installations directly occupy much less of the land within their boundaries. The environmental and social consequences of these divergent land occupancy patterns need further study.

Introduction

Here we estimate the power densities and capacity factors for wind and solar power plants with AC-

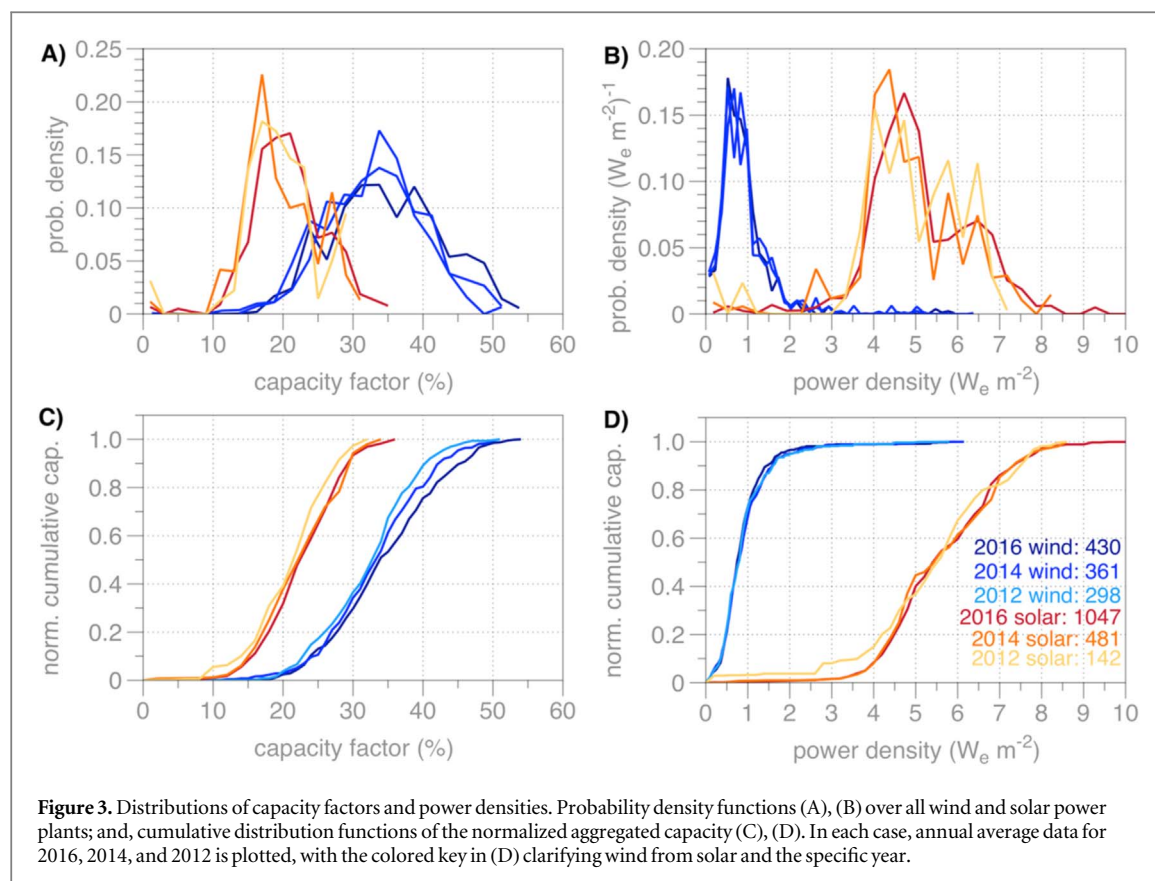


Table 1. Solar and wind power values for the various years, with average capacity factor and average power density weighted by the installed capacity.

Year	Solar power				Wind power		
	Installed capacity		Capacity factor (%)	Power density ($W_e m^{-2}$)	Installed capacity MW_i	Capacity factor (%)	Power density ($W_e m^{-2}$)
	MW_{dc}	MW_{ac}					
2010	78	67	22.70	5.92	18 868	30.55	0.93
2011	218	189	20.83	5.39	22 686	33.30	0.96
2012	833	700	21.09	5.37	27 096	32.60	0.93
2013	2032	1692	22.87	5.72	33 529	32.47	0.92
2014	3518	2865	23.04	5.63	34 389	33.67	0.94
2015	5157	4192	22.95	5.58	37 307	32.32	0.86
2016	8188	6612	23.52	5.70	45 400	34.94	0.90

capacities greater than 1 MW which generated electricity in the US from 1984–2016 installations. For wind we make a direct plant-by-plant bottom-up estimate while for solar our estimates of power density depend on a correlation analysis that provides a single estimate for the solar installed capacity density.

Data sources and methods

Following our Wind and Solar Methods (below), we computed annual averages from monthly generation (MWh/month) when 12 months of data is reported rather than using the data's annual averages (MWh yr^{-1}) which would obscure pre-startup or offline periods. Only about half of all wind and solar power plants were used in our analysis for the year

2016, with this ratio varying by technology and year. These solar and wind power plants were excluded because: (a) *Power Plants* could not be linked to *Electricity Generation* based on Plant Code, or (b) Capacity factors calculated from the *Power Plants* and *Electricity Generation* exceeded 100%, or (c) *Electricity Generation* was zero for any month in a given year, or finally (d) AC-capacities between *Power Plants* and *Detailed Data* differed by more than $\pm 10\%$.

These exclusions and filtering result in discrepancies between our dataset and those of the EIA (2018d). For 2016, the cumulative capacity of the wind power plants include in our data was 56% the EIA's estimate for total wind capacity while for solar capacity that figure was 44% (EIA 2018d). Our base *Power Plants* data collated power plants through early 2018,

but does not specify when the power plant came online, preventing capacity for 2016 from being quantified. *Detailed Data* provides nameplate capacity and month-year per power plant, but for 2016, total capacities are 109% and 150% the capacity for wind and solar respectively compared to (EIA 2018d). To verify that no region was systematically excluded, we spatially compared the raw EIA *Power Plant* locations (EIA 2018a) to those making it through our methodology, and found no obvious spatial gaps.

Wind methods

A detailed step-by-step methodology is described in the supplemental information, but broadly our approach for quantifying the area of US wind power plants begins with the location of the 57 636 wind turbines in the USWTDB (Hoen et al 2018). The USWTDB was reprojected to Contiguous USA Albers Equal Area (EPSG:102003) to enable precise small-area calculations across latitudes. Voroni polygons were calculated for each wind turbine using QGIS (2018). Using spatial linking, the Voroni polygons were linked to the *Power Plants* (EIA 2018a) and then filtered for an equivalent AC-installed capacity **within $\pm 10\%$** . The Plant Code in the *Power Plants* data was then used as the unique identifier for linking to *Electricity Generation* (EIA 2018a, 2018b) and *Detailed Data* (EIA 2018c). Capacity factors (MW_e/MW_i) of wind power plants are calculated from *Electricity Generation* and *Power Plants* (EIA 2018a, 2018c). Spatial and temporal curtailment by the grid operator was not included in this analysis, but will influence the results slightly (e.g. ERCOT region of Texas in 2009).

There is no well-established method to compute the area of each wind power plant. To do so, we compute a Voroni polygon (after Георгий Вороной) using QGIS (2018) for each wind turbine in the USWTDB which delineates the ground area that is closest to each individual turbine location compared to every other turbine. The Voroni polygon areas for wind turbines on the edge of wind power plants are very large, but the interior Voroni polygons are a useful quantification of the ground surface area per turbine. We compute the median Voroni polygon area for each wind farm **in the Contiguous US** and then estimate the area of the wind farm by multiplying this median Voroni polygon area by the number of wind turbines listed in the USWTDB (Hoen et al 2018).

These steps yield the wind power plant area (km^2), power density ($\text{W}_e \text{ m}^{-2}$), installed capacity density ($\text{MW}_i \text{ km}^{-2}$), and capacity factor for **430** wind power plants operating in 2016 (**45.4** GW_i).

Solar methods

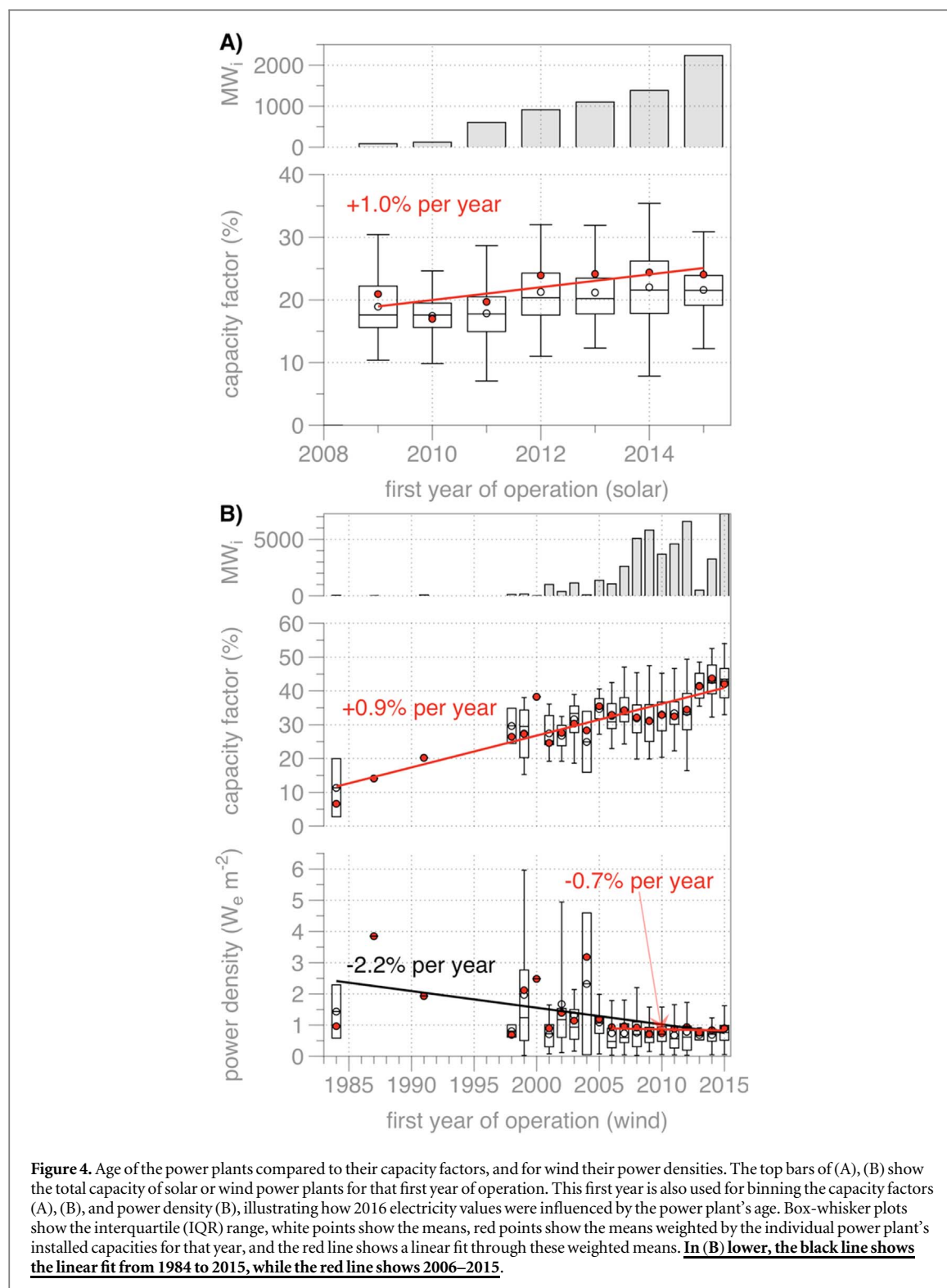
Our solar dataset begins with *Power Plants* (EIA 2018a). Using the unique Plant Code, we linked this file to *Electricity Generation* (EIA 2018b), resulting in 1 311 solar PV power plants. To reduce errors, we compare the installed capacity (MW_{ac}) values with the same Plant Code between *Power Plants* and *Detailed Data* (EIA 2018a), excluding the solar power plants that differ by $\pm 10\%$, leaving **1047** solar power plants for our 2016 analysis (**6.6** GW_{ac} , **8.1** GW_{dc}).

Results

Distributions of power densities and capacity factors are shown in figure 3. Considering capacity-weighted data for all power plants operational during 2016, the summary results are as follows. The mean and 90-percentile power densities for wind are **0.90** and **1.48** $\text{W}_e \text{ m}^{-2}$, while the corresponding values for solar are **5.7** and **7.5** $\text{W}_e \text{ m}^{-2}$. Note that systematic uncertainty in the distribution of power densities are significantly larger for solar than for wind because the solar power results are derived from a fixed estimate of capacity density, whereas the wind results are computed directly. Our mean and 90-percentile capacity factors are **34.9%** and **46.0%** for wind, while the corresponding values for solar are **23.5%** and **30.0%**. Note that the capacity factors from EIA for 2016 are 34.5% for wind and 25.1% for solar (EIA 2018d), and we expect that the discrepancy arises from the data sampling issues discussed above. Solar and wind power installed capacities, power densities, and capacity factors from 2010 to 2016 are shown in table 1.

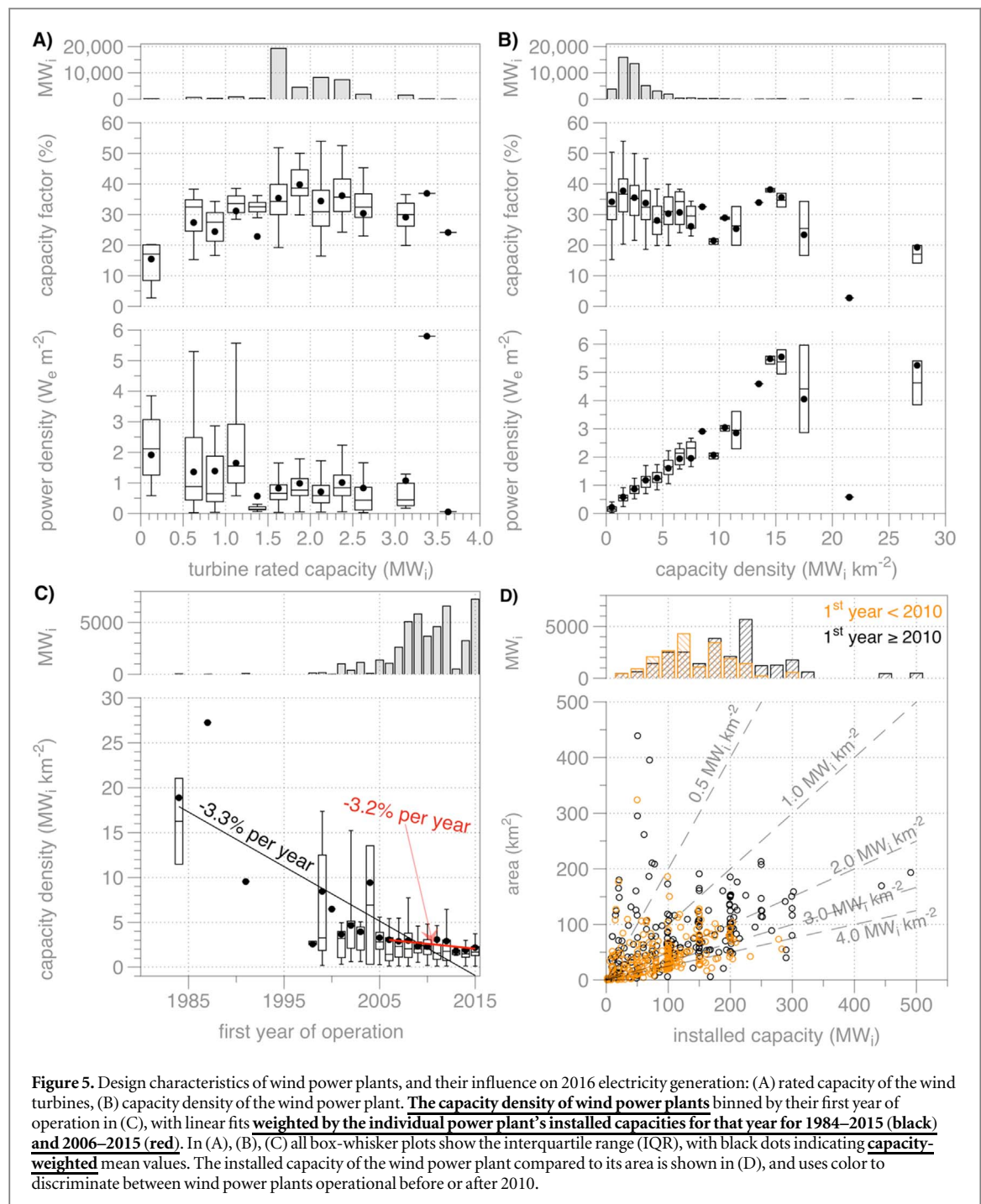
Capacity factors for wind power have increased by **0.9%** per year over the years 1984–2015 (figure 4(B)). The increase in wind's capacity factor is particularly evident this decade. Wind farms operating since 2010 have a mean capacity factor of **37.3%** for 2010–2016, whereas the capacity factor from 1984 to 2009 is **31.6%**.

There is no significant trend in the power density of wind power plants. This result is surprising given the increase in capacity factor. What underlies it? Wind power plants have three defining characteristics: the rated capacity of individual turbines, the installed capacity density of the wind farm, and the area of the wind farm. The capacity factor and power density of the wind power plants show no relationship to the rated capacity of the individual wind turbines (figure 5(A)), whereas capacity factor and power density do vary with capacity density (figure 5(B)). Note that the highest power densities are achieved with the highest capacity densities, but the highest capacity factors are achieved with the lowest capacity densities. **Based on their first year of operation, the capacity density of wind power plants has decreased by 3.3% per year since 1984, or 3.2% per year over the last**



10 years (figure 5(C)). The capacity density peaked at about $5 \text{ MW}_i \text{ km}^{-2}$ for turbines installed between 2002 and 2005, and has since decreased to about $2.0 \text{ MW}_i \text{ km}^{-2}$. **Overall**, the average installed capacity density of all wind farms was $2.7 \text{ MW}_i \text{ km}^{-2}$ (figure 5(D)). In summary, we find that while improved wind turbine design and siting have increased capacity factors (and greatly reduced costs) they have not altered power densities.

Finally, we examined the relationship between power plant area and power density. For solar, there is no clear relationship between area and power density (figure 7(A)), whereas for wind, there is a strong relationship (figure 7(B)). While many wind power plants with areas less than 20 km^2 generate more than $1.0 \text{ W}_e \text{ m}^{-2}$, power density decreases with increasing power plant size. This result was previously observed for $0\text{--}20 \text{ km}^2$ wind power plants by (MacKay 2013). We



verify this early result, and extend it by showing that wind's power density reaches an asymptote of about $0.50 \text{ W}_e \text{ m}^{-2}$ when the wind plant area exceeds about 150 km^2 . **Note that we found power densities of about $0.1 \text{ W}_e \text{ m}^{-2}$ for the largest wind-farms in our study, which had areas of $250\text{--}450 \text{ km}^2$. While these plants passed all the quality-checks of the Wind Method, we feel they are not representative. While these plants have comparable capacity factors (24%–47%), they have very low capacity densities ($0.11\text{--}0.23 \text{ MW}_i \text{ km}^{-2}$). These plants are, in general, isolated from neighboring plants and/or have turbines placed along ridgelines, which is problematic for quantifying a representative median Voroni**

polygon area. We therefore judge that the power densities of about $0.50 \text{ W}_e \text{ m}^{-2}$ in figure 7(B), which we find for plants of $100\text{--}200 \text{ km}^2$, as most illustrative to what would likely be achieved by replicating existing wind plants in adjacent regions with similar wind resources.

Discussion

Solar's mean power density in 2016 was $5.7 \text{ W}_e \text{ m}^{-2}$. Our approach for estimating the area of solar farms is not fully bottom-up so this estimate is subject to systematic error. It is possible, for example, that

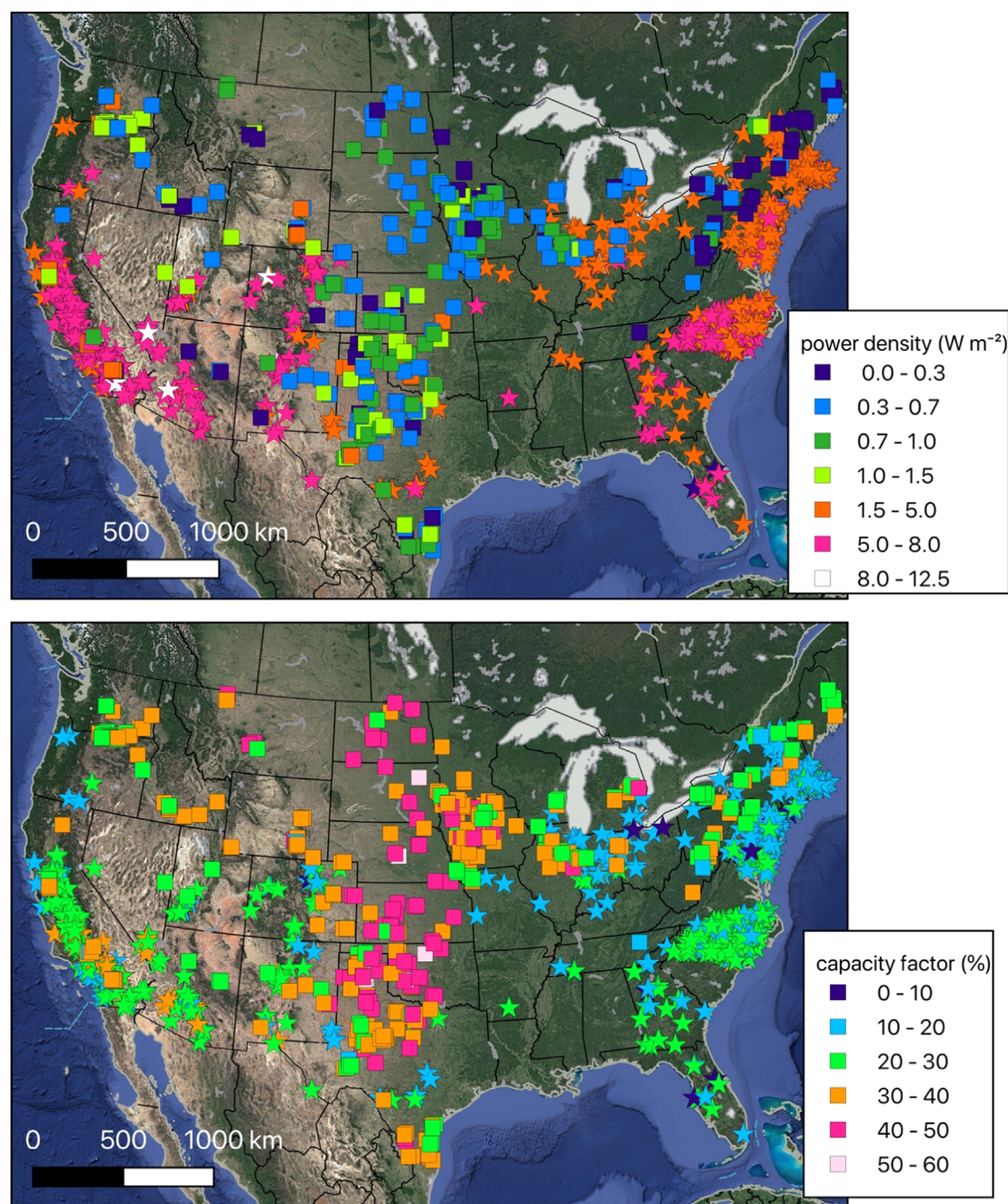


Figure 6. Spatial distribution of power density (top) and capacity factor (bottom) for 2016. Wind power plants represented as squares and solar power plants as stars.

capacity densities have changed significantly given that the data used in our analysis is about 5 years old. That said, the assumption by (Jacobson *et al* 2018) that urban rooftops can be retrofitted with a capacity density 4.5 times higher than the commercial-scale solar plants measured by (Ong *et al* 2013) seems highly unlikely, as does the resulting $24\text{--}27 \text{ W m}^{-2}$ power density (Jacobson *et al* 2018). It is also possible that capacity densities vary strongly with larger size installations (see figure 2(A)). However, given that our analysis finds only a very weak relationship between module efficiency or installation size and capacity density, we expect the errors are small, likely less than 20%.

Wind's mean power density in 2016 was 0.90 W m^{-2} . This observed mean is consistent with estimates based on atmospheric theory and modeling (Gustavson 1979, Keith *et al* 2004, Wang and Prinn 2010, Miller *et al* 2011, Gans *et al* 2012, Jacobson and Archer 2012, Marvel *et al* 2012, Adams and Keith 2013, Miller *et al* 2015, Miller and Kleidon 2016) which predicted that large-scale wind power densities would be under 1.0 W m^{-2} and also that power densities will decrease with increasing size of the wind farm installation. This observed mean power density is much smaller than many common estimates (Archer and Jacobson 2005, Lu *et al* 2009, Sta. Maria and Jacobson 2009, Jacobson and Delucchi 2011, Lopez

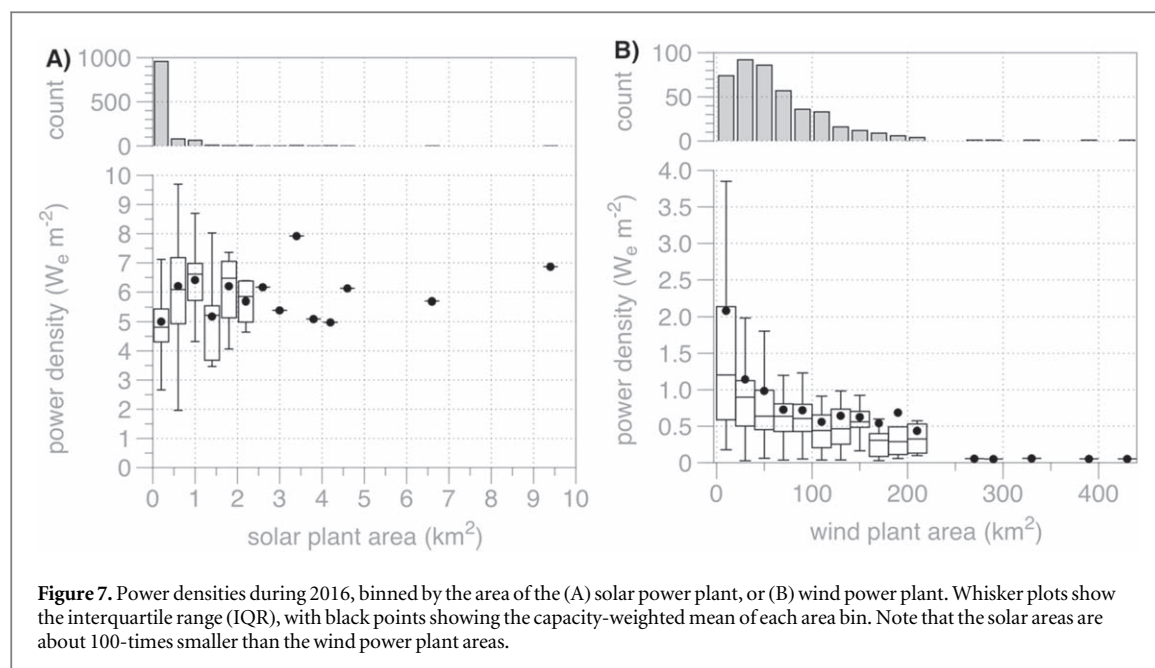


Figure 7. Power densities during 2016, binned by the area of the (A) solar power plant, or (B) wind power plant. Whisker plots show the interquartile range (IQR), with black points showing the capacity-weighted mean of each area bin. Note that the solar areas are about 100-times smaller than the wind power plant areas.

et al 2012, US DOE 2014, World Bank Fund 2018). Examples include 227 W m^{-2} over the windiest 10% of global land (World Bank Fund 2018), $3.3 \text{ W}_e \text{ m}^{-2}$ over the entire Earth's surface (Jacobson and Delucchi 2011), $1.7 \text{ W}_e \text{ m}^{-2}$ over about 1/3 of the Continental US (Lopez *et al* 2012), or $1.4 \text{ W}_e \text{ m}^{-2}$ over about 2% of the US with excellent wind resources (US DOE 2014). These are also by no means the highest estimates in the literature. For example, (World Bank Fund 2018) quantify a wind power density of 808 W m^{-2} over the windiest 10% of US land, and (Kammen and Sunter 2016) estimated an upper bound of $35 \text{ W}_e \text{ m}^{-2}$ for wind power at urban-scales based on a study observing numerous vertical axis turbines which generated up to $47 \text{ W}_e \text{ m}^{-2}$ over an area of about 50 m^2 (Dabiri 2011).

There are two main reasons for these discrepancies in wind power density. First, many estimates did not account the interactions between wind turbine arrays and the atmospheric boundary layer. The limit to large-scale wind power density is the downward flux of kinetic energy from the free troposphere, a **global** value that is about 1 W m^{-2} (Lorenz 1955, Peixoto and Oort 1992, Kim and Kim 2013). The effect of this atmospheric limit is illustrated by the relationship between wind power plant's area and power density. Second, many studies assume installed capacity densities which are too high. While we observed an average capacity density of $2.7 \text{ MW}_i \text{ km}^{-2}$ **with about $2.0 \text{ MW}_i \text{ km}^{-2}$ common to 2013–2015 installations**, (Rinne *et al* 2018) assume $5.5\text{--}9.4 \text{ MW}_i \text{ km}^{-2}$, (Jacobson *et al* 2018) assume $7.2 \text{ MW}_i \text{ km}^{-2}$, (Lopez *et al* 2012) assumed $5.0 \text{ MW}_i \text{ km}^{-2}$, the US-DOE *Wind Vision: A New Era for Wind Power in the United States* (US DOE 2014) assumed $3.0 \text{ MW}_i \text{ km}^{-2}$. By assuming 2–5 times the observed capacity density but ignoring the atmospheric limits, these estimates

resulted in power densities that are 2–5 times higher than observations.

Given that larger wind power plants have smaller power densities and given that a major increase in total wind power generation will presumably require expanding wind power plants into less-than-ideal locations, it seems likely that wind power density will decrease with time. It therefore seems—contrary to many prior estimates—unlikely that the power densities of greater than $1 \text{ W}_e \text{ m}^{-2}$ will be realized over substantial areas, and likely that average power densities **of about $0.5 \text{ W}_e \text{ m}^{-2}$ will become increasingly common**.

As an example of the implications of these results, consider Germany and its ambitious energy transformation policy (Energiewende). Germany's primary energy consumption rate is 1.28 W m^{-2} (BP 2018). If **the average** US wind power density of $0.90 \text{ W}_e \text{ m}^{-2}$ was applicable to Germany, then devoting all German land to wind power would meet about **70%** of Germany's total primary energy consumption, while if German wind power performs like the best 10% of US wind ($1.48 \text{ W}_e \text{ m}^{-2}$), then generation would be **115%** of Germany's consumption. Finally, if Germany's goal was to generate the most wind power without economic constraints, very high capacity densities (e.g. $10 \text{ MW}_i \text{ km}^{-2}$) could be deployed, reducing capacity factors but possibly raising the power density to $2.0 \text{ W}_e \text{ m}^{-2}$ and meeting **135%** of consumption. Whereas for solar at $5.7 \text{ W}_e \text{ m}^{-2}$, **22%** of Germany's land area would need to be devoted to commercial-scale solar to meet total primary energy consumption.

Power densities clearly carry implications for land use. Meeting present-day US electricity consumption, for example, would require 12% of the Continental US land area for wind at $0.5 \text{ W}_e \text{ m}^{-2}$, or 1% for solar at $5.7 \text{ W}_e \text{ m}^{-2}$. US electricity consumption is just 1/6

total primary energy consumption (BP 2018), so meeting total consumption would therefore require 72% and 6% respectively for US wind and solar. Of course, like the Germany example, no single energy source is likely to ever supply all electric power. These comparisons nevertheless provide a benchmark for understanding the implications of power densities for land use, while recognizing that solar and wind power also occupy the area within the power plant boundary differently. These observation-based results should be considered in light of the fact that (a) decarbonizing the energy system will require considerably more primary power than current electricity demand, (b) demand may continue to grow, and finally, (c) that many areas of the world have higher energy demand per unit area than does the Continental US.

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